

Ashwin Swamy

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EDUCATION

King's College School, Wimbledon

London, United Kingdom

Year 13 (Grade 12) Senior School

2022-Present

Studying the IBDP: Year 13 (Predicted): 45/45

- Higher Level: Mathematics AA, Physics, Chemistry, and Standard Level: Philosophy, Chinese, English
- Achieved L6th Physics Prize: 1st of 100+ physics cohort; Top of year in Year 12 IB exams (7s in all)

(I)GCSEs: 11 9s (One of 116 students in UK) with 100% in Further Mathematics

- Mathematics, Further Mathematics, Physics, Chemistry, Biology, Computer Science, Geography, English Language, English Literature, Mandarin, French

SAT: 1550, English: 760, Maths: 790

RESEARCH & PROJECTS

Payload-Responsive EV Chassis Research

2025-Present

Accelo Mobility, SINE IIT Bombay-backed EV Startup

- Identified engineering gap in commercial EV platforms: ICE-retrofitted rigid platforms are incompatible with the dynamic payload variations of last-mile delivery, causing instability, accelerated wear and risk of battery fires
- Designed an optimised semi-spaceframe chassis through CAD and 100+ FEA iterations spanning possible load conditions
- Traded reinforcement against weld access and material cost to ensure manufacturability and affordable access in emerging markets; increased safety factor by 4.6x compared to ladder-frame baseline
- Engaged with manufacturing partner Accelo Mobility to run my design in a prototype EV delivery pickup
- Published in National High School Journal of Science, Flagship Journal (~1% acceptance)
- Presented at Sigma Xi Student Research Conference

UK CanSat Competition - StratoSat

2025-2026

ESA/ESERO UK

- Project Manager and Recovery System Lead for StratoSat - scored 100% (PDR) and 32/34 (CDR); past UK regionals
- Led project planning, risk analysis, and authored design review documentation submitted to ESA/ESERO judges
- Designed and built the recovery parachute from first principles: sized canopy area, measured opening shock loads
- Contributed to CAD and 3D printing of the PLA structure

Independent Economics Research: Decentralised Finance & Payment Infrastructure

2025-2026

- Argued that a cashless economy's risks depend on who controls the infrastructure, comparing payment systems in India, China, Nigeria and Sweden
- Extended the analysis to stablecoins: how money moving outside banks could weaken monetary policy and dollarise emerging-market deposits
- Shortlisted for Economics at 2026 John Locke Essay Competition

EXPERIENCE

Tech Mahindra

London, United Kingdom

Internship – Platform economics client project for Bentley Motors

July 2026

- Traced a 42% fall in Bentley's operating profit to have been influenced by the cancellation of a shared chassis platform
- Analysed how shared platforms fix suspension hard points & standardise aesthetics, constraining Bentley's core source of customer value
- Benchmarked Bentley platform strategy against peer OEMs; authored client report proposing volume-linked platform charging

REGENT Craft & Defense

Rhode Island, USA

Summer Work Experience

July 2025

- Shadowed Electrical Engineer at an electric sea-glider company
- Gained hands-on experience installing DC-DC converter system in prototype sea-glider powering avionics systems

LEADERSHIP & ACTIVITIES

Cricket4Life – Co-Founder

2022-Present

Roehampton Cricket Club

- Co-founded integration initiative for Afghan refugee teammates after end of state support dispersed players across the UK
- Designed and ran weekly English programme & worksheets built on Level 1 English qualification syllabus needed to coach
- 2 teammates now employed as paid club cricket coaches
- Co-led fundraising across 2 schools, raising £1,000+ & cricket kit for teammates' coaching sessions and masterclasses

IGNITE Initiative - Founder
King's College School, Wimbledon

- Founded design & innovation initiative connecting students with engineering industry experts & demystifying STEM career paths
- Directed session with BP Senior VP of Technology and Innovation, working with Audi F1, for 50 students across 4 year groups
- Built confirmed line-up of 3 industry experts for speaker series after original speaker withdrew
- Building international product design competition across partner schools in Bangkok, Monaco and the UK, judged by experts

2025-Present

Mirror Board - Member | Aerospace & Aviation Society - Co-Founder
King's College School, Wimbledon

- Co-founded Aerospace & Aviation Society after spotting unmet demand for Year 7-8; ran 13 sessions for 15 pupils
- Selected as 1 of 14 from 231 students to Mirror Board, the school's highest student governance body, chaired by the Headmistress
- Drew on society's launch struggles to diagnose a failing clubs system for younger years
- Passed reform for 150 clubs with Head of Co-curricular: simpler formation route, digitalised advertising for younger year groups & club awards

2024-Present

AWARDS & CERTIFICATIONS

• BPhO Senior Physics Challenge - Gold Award	2026
• UKMT Senior Mathematical Challenge - Gold Award, qualified for Senior Kangaroo	2025
• John Locke Essay Competition - Shortlisted (Economics)	2026
• John Locke Essay Competition - Finalist (Junior)	2023
• British Science Association – CREST Gold Award	2026
• British Taekwondo National Championship – Team Silver Medal	2024
• ABRSM Piano Grade 7 - Distinction	2024

SKILLS AND INTERESTS

Languages: English (Native), French (Proficient), Mandarin (Proficient), Tamil (Mother Tongue)

Technical: CAD & FEA (Fusion360), Python, JavaScript/HTML, graphic design & video editing (DaVinci Resolve)

Interests: National-level taekwondo (silver medalist), music producer, rock band member, F1 enthusiast